$$\log_5(2x-6)-\log_5 2=\log_5 3$$

$$\log_5\frac{2x-6}{2}=\log_5 3$$

$$\frac{2x-6}{2}=3$$

$$2x-6=6$$

$$2x=12$$

$$x=6$$

$$T=2\pi=360^0$$

$$\sin(\alpha\pm360)=\sin\alpha$$

$$\cos(\alpha\pm360)=\cos\alpha$$

$$tg(\alpha\pm\pi)=tg\,\alpha$$

$$\sin(\alpha \pm \beta) = \underline{\sin\alpha\cos\beta} \pm \underline{\sin\beta\cos\alpha}$$

$$\sin(\underbrace{30+60}_{=90}) = \sin 30 \cos 60 + \sin 60 \cos 30$$

$$\frac{1}{2}\cdot\frac{1}{2} + \frac{\sqrt{3}}{2}\cdot\frac{\sqrt{3}}{2} =$$

$$\frac{1}{4} + \frac{3}{4} = \frac{4}{4} = 1$$

$$\cos(\alpha \mp \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$

$$\sin(135^\circ) = \sin(90+45) =$$

$$\underbrace{\sin 90}_{1}\ \underbrace{\cos 45}_{\frac{\sqrt{2}}{2}} + \underbrace{\sin 45}_{\frac{\sqrt{2}}{2}}\ \underbrace{\cos 90}_{0}$$

$$\frac{\sqrt{2}}{2} + 0 = \frac{\sqrt{2}}{2}$$

$$\cos(135^\circ) = \underbrace{\cos 90}_{0}\ \underbrace{\cos 45}_{\frac{\sqrt{2}}{2}} - \underbrace{\sin 90}_{1}\ \underbrace{\sin 45}_{\frac{\sqrt{2}}{2}}$$

$$= -\frac{\sqrt{2}}{2}$$

$$\cos(150^\circ) = \cos(90+60) =$$

$\underbrace{\cos 90 \cos 60}_{0} - \underbrace{\sin 90}_{1} \underbrace{\sin 60}_{\frac{\sqrt{3}}{2}}$

$0 \quad -1 \cdot \frac{\sqrt{3}}{2} = -\frac{\sqrt{3}}{2}$

$\sin(150^\circ) = \sin 90.$

$\sin(90+60) = \sin 90 \cdot \cos 60 + \sin 60 \cdot \cos 90$

$= 1 \cdot \frac{1}{2} + \frac{\sqrt{3}}{2} \cdot 0 = \frac{1}{2}$

$\sin(120) = \underbrace{\sin 90}_{1} \cdot \underbrace{\cos 30}_{\frac{\sqrt{3}}{2}} + \underbrace{\sin 30}_{\frac{1}{2}} \underbrace{\cos 90}_{0}$

$= \frac{\sqrt{3}}{2}$

$\cos(120) = -\frac{1}{2}$

$12 \sin 150 \cdot \cos 120 =$

$$\boxed{\begin{aligned} \sin(\alpha \pm \beta) &= \sin\alpha\cos\beta \pm \sin\beta\cos\alpha \\ \cos(\alpha \pm \beta) &= \cos\alpha\cos\beta \mp \sin\alpha\sin\beta \end{aligned}}$$

$$\sin 150 = \sin 90 \cos 60 + \sin 60 \cos 90 = \frac{1}{2}$$

$$\cos 120 = -\frac{1}{2}$$

$$12 \cdot \frac{1}{2} \cdot \left(-\frac{1}{2}\right) = -3$$

$$\operatorname{tg}(\alpha \pm \beta) = \frac{\sin(\alpha \pm \beta)}{\cos(\alpha \pm \beta)}$$

$$\operatorname{tg}(120^\circ) = \frac{\sin(120)}{\cos(120)} \ldots$$

$$\operatorname{tg} 75^\circ = \frac{\frac{\sqrt{2}}{2}\frac{\sqrt{3}}{2} + \frac{1}{2}\cdot\frac{\sqrt{2}}{2} = \frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4}}{\frac{\sqrt{6}}{4} + \frac{\sqrt{2}}{4}}$$

$$17 \cdot tg\,13 \cdot tg\,77$$

$$13 + 77 = 90$$

$$tg\,13 = tg(90 - 77)$$

$$tg(90-77) = \frac{\sin(90-77)}{\cos(90-77)} =$$

$$= \frac{\overset{1}{\cancel{\sin 90}}\cos 77 - \sin 77 \overset{0}{\cancel{\cos 90}}}{\underset{0}{\cancel{\cos 90}}\cos 77 + \underset{1}{\cancel{\sin 90}}\sin 77}$$

$$\frac{\cos 77}{\sin 77}$$

$$17 \cdot \frac{\cancel{\cos 77}}{\cancel{\sin 77}} \cdot \frac{\cancel{\sin 77}}{\cancel{\cos 77}} = \underline{\underline{17}}$$

$$5\ tg\,17 \cdot tg\,107$$

$$107 = 90 + 17$$

$$\operatorname{tg}(90+17) = \frac{\sin(90+17)}{\underbrace{\cos(90+17)}} = \frac{\cos 17}{-\sin 17}$$

$$\cos 90 \cos 17 \underbrace{- \sin 90 \sin 17}$$

$$5 \cdot \operatorname{tg} 17 \cdot \frac{\cos 17}{-\sin 17} = 5 \cdot \frac{\cancel{\sin 17}}{\cancel{\cos 17}} \left(\frac{\cancel{\cos 17}}{-\cancel{\sin 17}} \right)$$

$$\frac{5}{-1} = -5$$

$$59 \operatorname{tg} 56 \cdot \operatorname{tg} 34$$

$$56 + 34 = 90$$

$$\operatorname{tg} 56 = \operatorname{tg}(90-34) = \frac{\cos 34}{\underline{\underline{+\sin 34}}} ?$$

$$\cos(90-34) = \cos 90 \cos 34 + \sin 90 \sin 34$$

$$59\,\frac{\cos 34}{\sin 34}\cdot\frac{\sin 34}{\cos 34} = 59$$

$$\cos(-750) = \cos(-750+720)$$

$$= \cos(-30^\circ) = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\boxed{\begin{aligned}\cos(-\alpha) &= \cos\alpha \\ \sin(-\alpha) &= -\sin\alpha\end{aligned}}$$

$$\sin(-30) = -\sin 30^\circ = -\frac{1}{2}$$

$$-4\sqrt{3}\cos(-930)$$

$$-930+360 = -570$$

$$\underbrace{-570}+\underbrace{360} = -210$$

$$-210+360 = 150$$

$-4\sqrt{3}\cos 150$

$20\sqrt{3}\cos 390 = 20\sqrt{3}\cos 30 =$

$20\sqrt{3}\cdot\frac{\sqrt{3}}{2} = \frac{\overset{10}{\cancel{20}}\cdot\sqrt{9}}{\cancel{2}}$